b) A...
ACROSS ALL SUBNETS AND CONNECTS THE SUBNET THROUGH A ROUTER

AIM:
To assign static IP addresses to all devices in different subnets and establish communication between the subnet through a router

STEP:-
1. Place a router in the network topology
2. Connect the router to the switch using appropriates cables
3. Configure the router interfaces.
4. Assign an IP address and subnet mask to each other route interface
5. Configure the switches.
6. Assign static IP address to all computers in every subnet
7. Configure the subnet mask for each devices
8. Assigns the appropriate default by gateway to each computer
9. Verify the IP address configuration of all devices
10. Save the router configuration

Router1

Physical Config CLI

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
to up
exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.224
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state
to up
exit
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     192.168.2.1     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

Copy Paste

PC0

Physical Config Desktop Custom Interface

Command Prompt

```
PC>ping 192.168.1.12

Pinging 192.168.1.12 with 32 bytes of data:

Reply from 192.168.1.12: bytes=32 time=0ms TTL=128
Reply from 192.168.1.12: bytes=32 time=0ms TTL=128
Reply from 192.168.1.12: bytes=32 time=0ms TTL=128
Reply from 192.168.1.12: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

PC>ping 192.168.2.11

Pinging 192.168.2.11 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.11: bytes=32 time=1ms TTL=127
Reply from 192.168.2.11: bytes=32 time=0ms TTL=127
Reply from 192.168.2.11: bytes=32 time=6ms TTL=127

Ping statistics for 192.168.2.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 6ms, Average = 2ms

PC>
```

11. Check the network topology for errors
12. Verify that all devices are properly connected

RESULT:-

The, static IP address were successfully assigned to all devices, and communication between different subnet was established through the router.